

QX4001 – Molecules to Materials

Rotation 1 – Question Set 1

- 1) Define “orbital” in a single short sentence. In your group you may come up with more than one definition. Be prepared to rank them according to suitability (better to worse) and explain the reasons for your ranking.
- 2) What do you think of the term “orbital”? Is there any other word that is similar in English? Why do you think it was proposed originally? Can you think of any issues in student understanding caused by the use of the word “orbital”?
- 3) How strict are chemists in their use of Orbital Theory? In other words: what’s the use of Orbital Theory in understanding chemistry?
- 4) Write electronic configurations for elements $Z=37$ and $Z=53$. What type of bond do you expect them to form and why?
- 5) Write electronic configurations for Li^+ , Cl^- , Fe^{+++} .
- 6) Write all possible quantum number sets for electrons on the following orbitals: 4s, 3p, 4d.
- 7) What possible elements have an electron with quantum numbers $[3, 1, 0, 1/2]$ as their highest energy filled (or half filled) orbital? What possible elements have an electron with quantum numbers $[3, 0, 0, 1/2]$ as their highest energy filled orbital?
- 8) Given that $h = 6.63 \times 10^{-34} \text{ Js}$ and $c = 3.00 \times 10^8 \text{ ms}^{-1}$ Calculate the wavelength (in nm) of light with energy $6.54 \times 10^{-19} \text{ J}$ per photon. In what region of the electromagnetic spectrum does this light fall?
- 9) Write out the electronic configuration for the following elements: Co and Se.
- 10) Draw representations of the following atomic orbitals: 1s, 2s, 2p, 3p and 3d. Discuss what orbital characteristics are relevant in ensuring that your diagram is complete and truly representative of these orbitals. Provide definitions for these characteristics. (If you are unsure about “orbital characteristics” compare 1s with 2p orbitals: in what way do these two differ? These are “orbital characteristics”).